

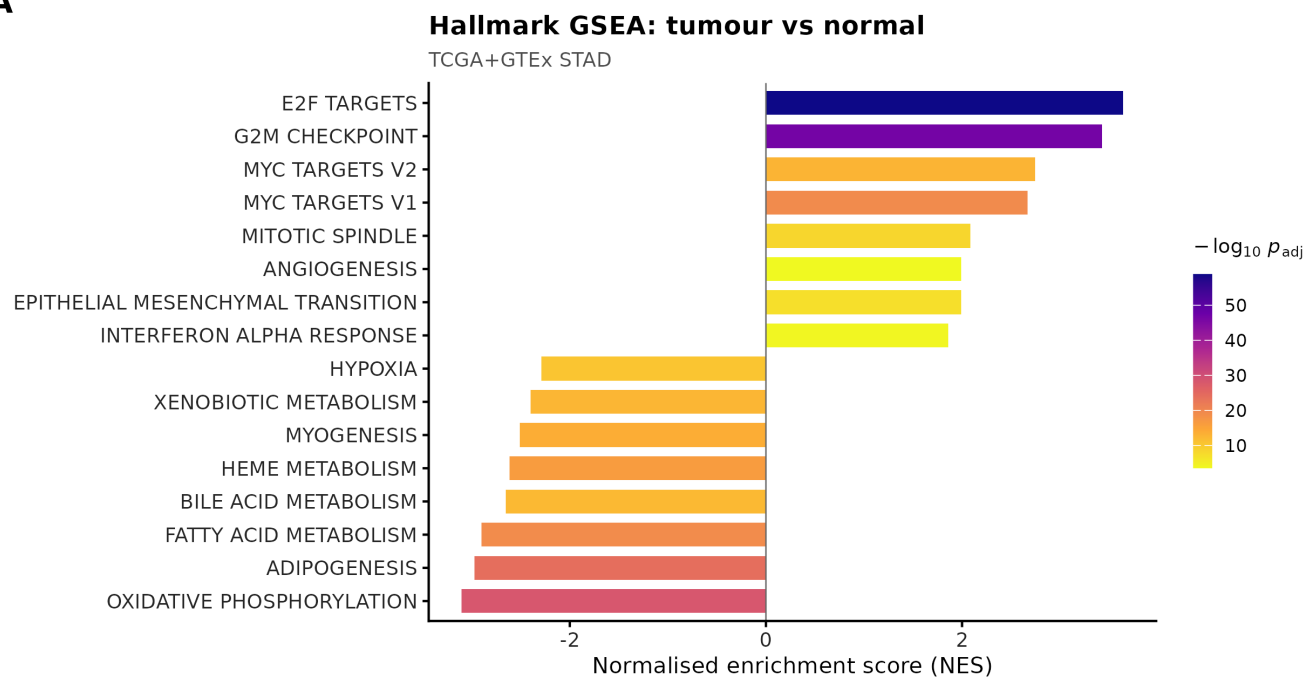
Gastric Cancer Multi-Omics

Complete figure set — every image standalone, one per page

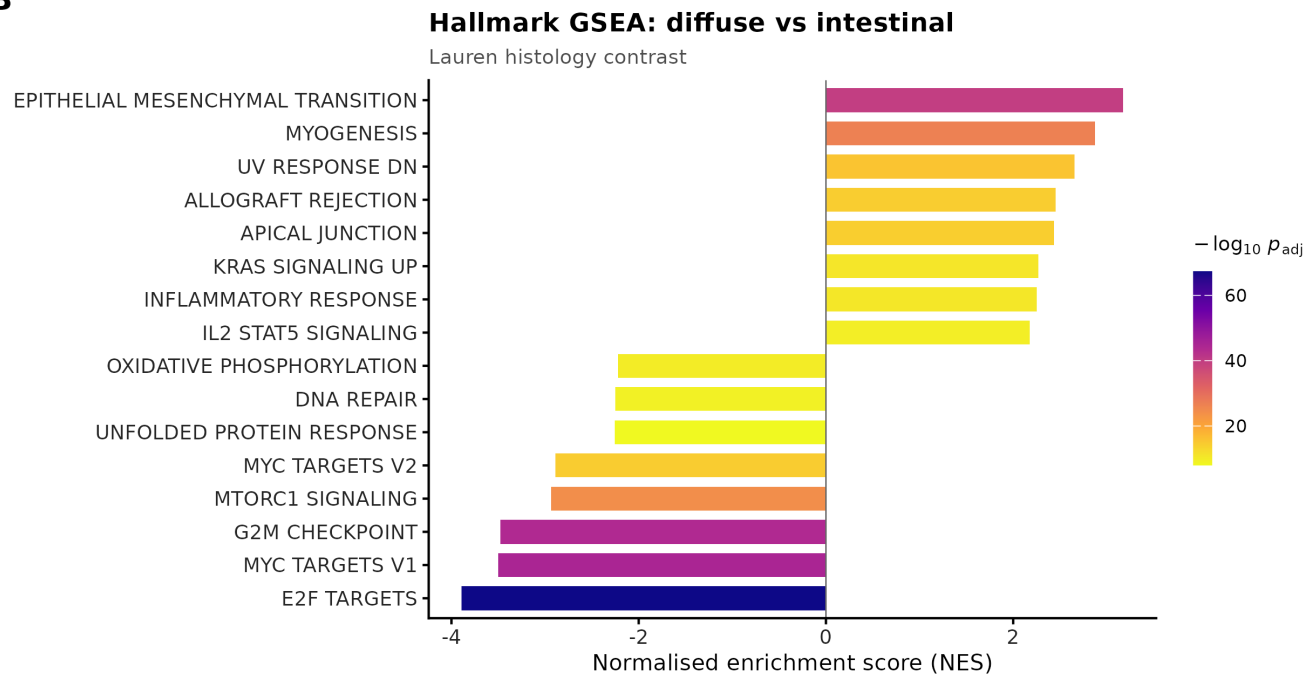
28 figures · generated 02 August 2026
each reproduced from its committed source file

Figure 1

A



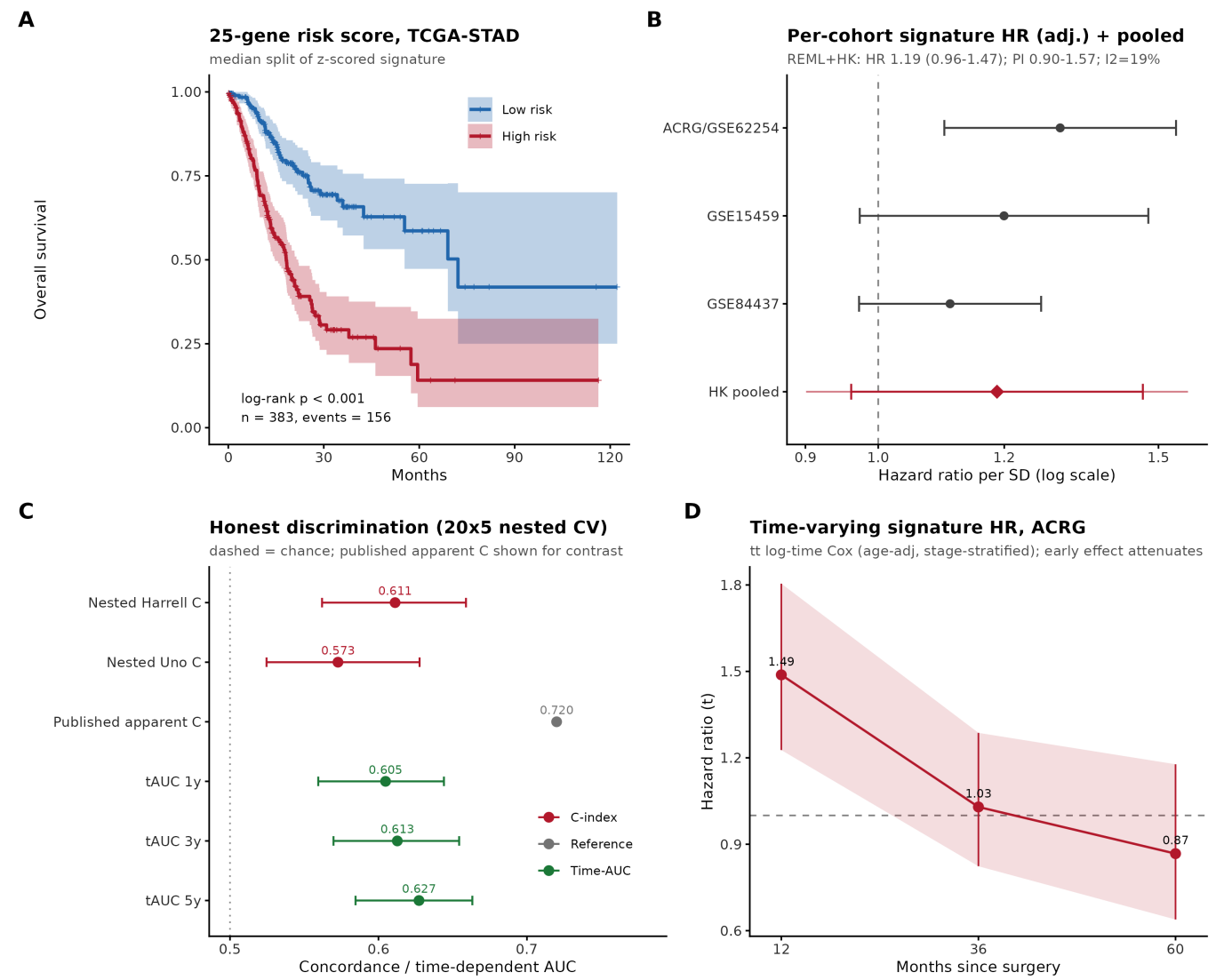
B



Tumour transcriptional programmes.

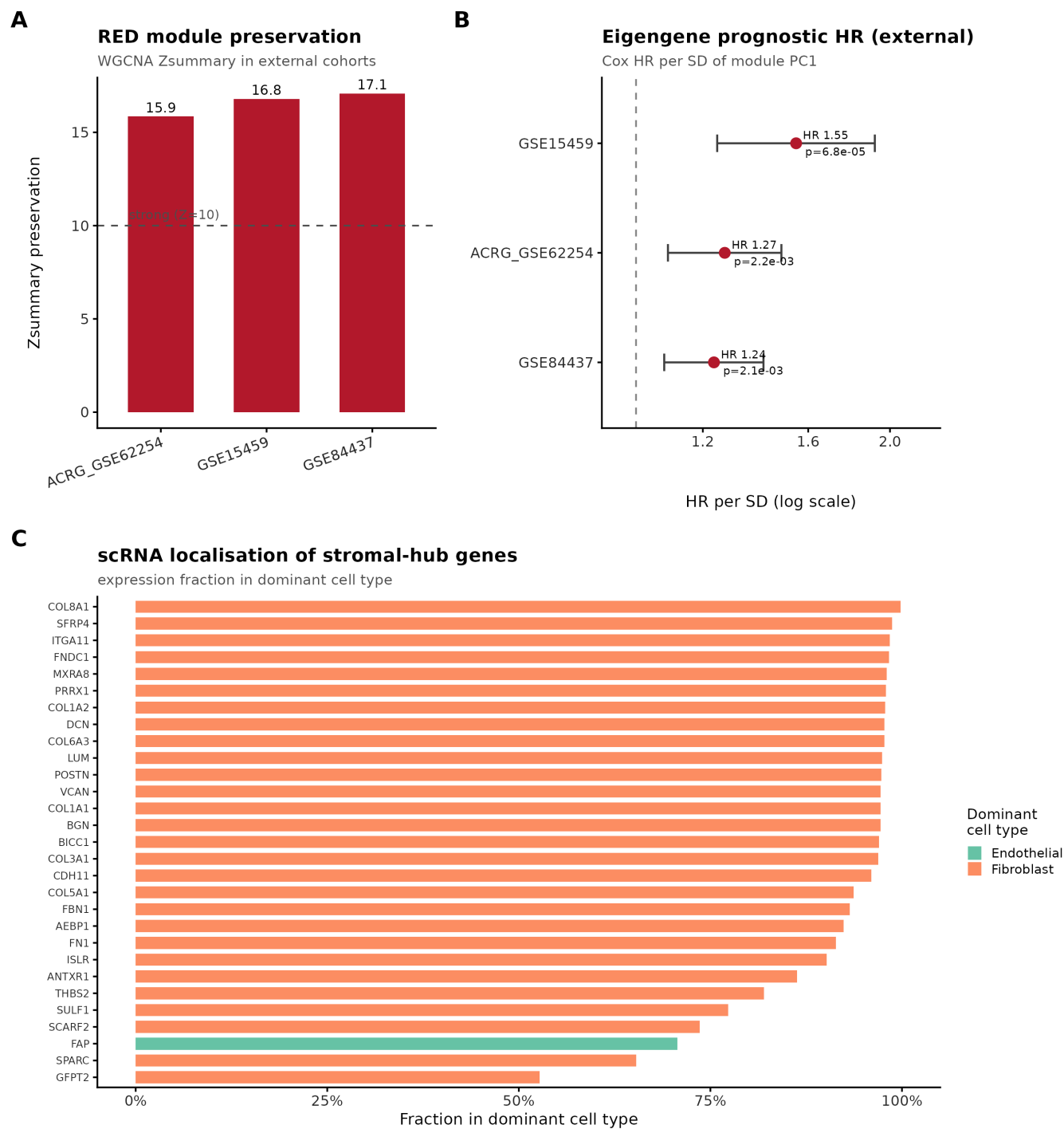
results/figures/Fig1.png

Figure 2



The 25-gene signature, reported honestly.
results/figures/Fig2.png

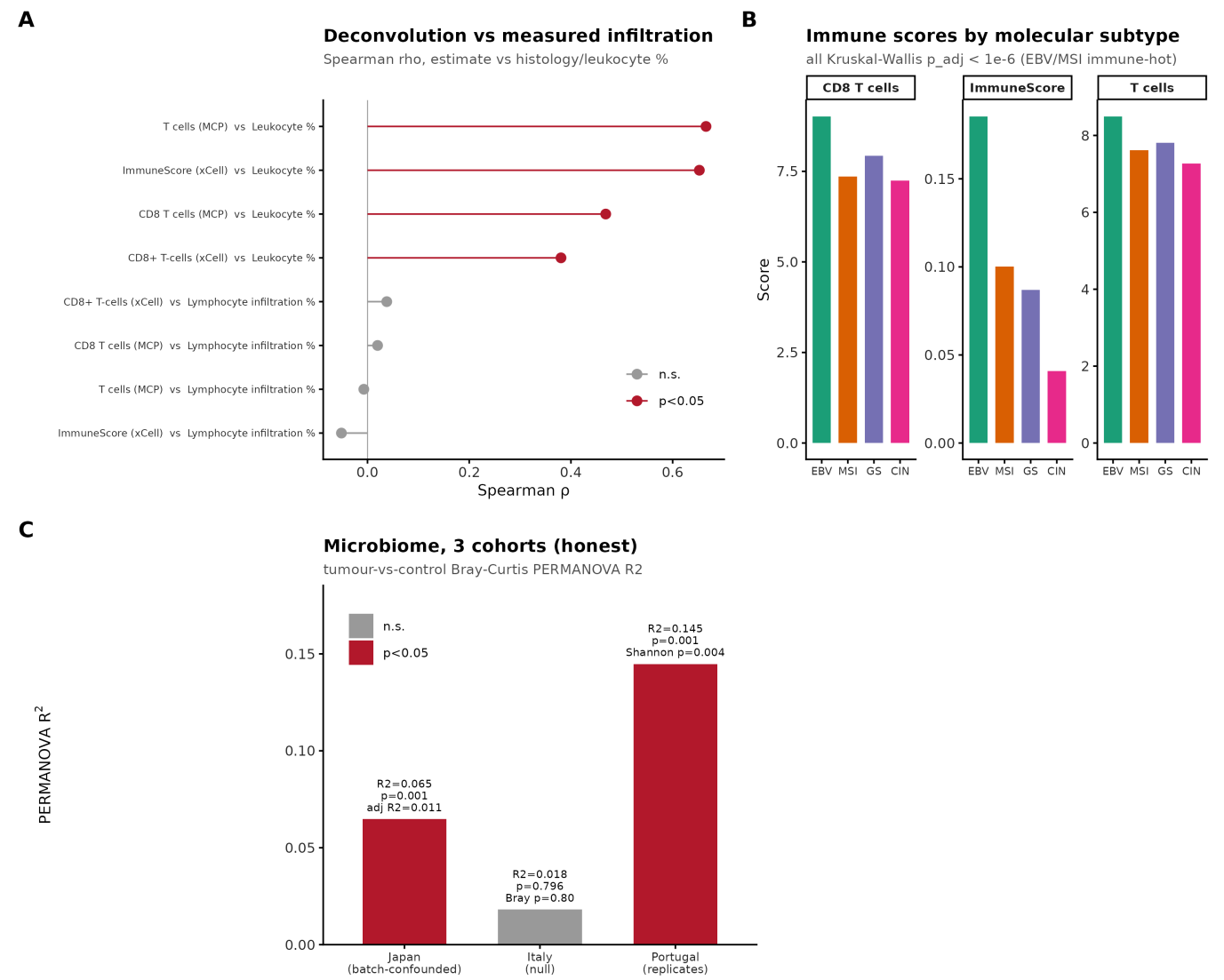
Figure 3



The externally-validated fibroblast/stromal module (primary finding).

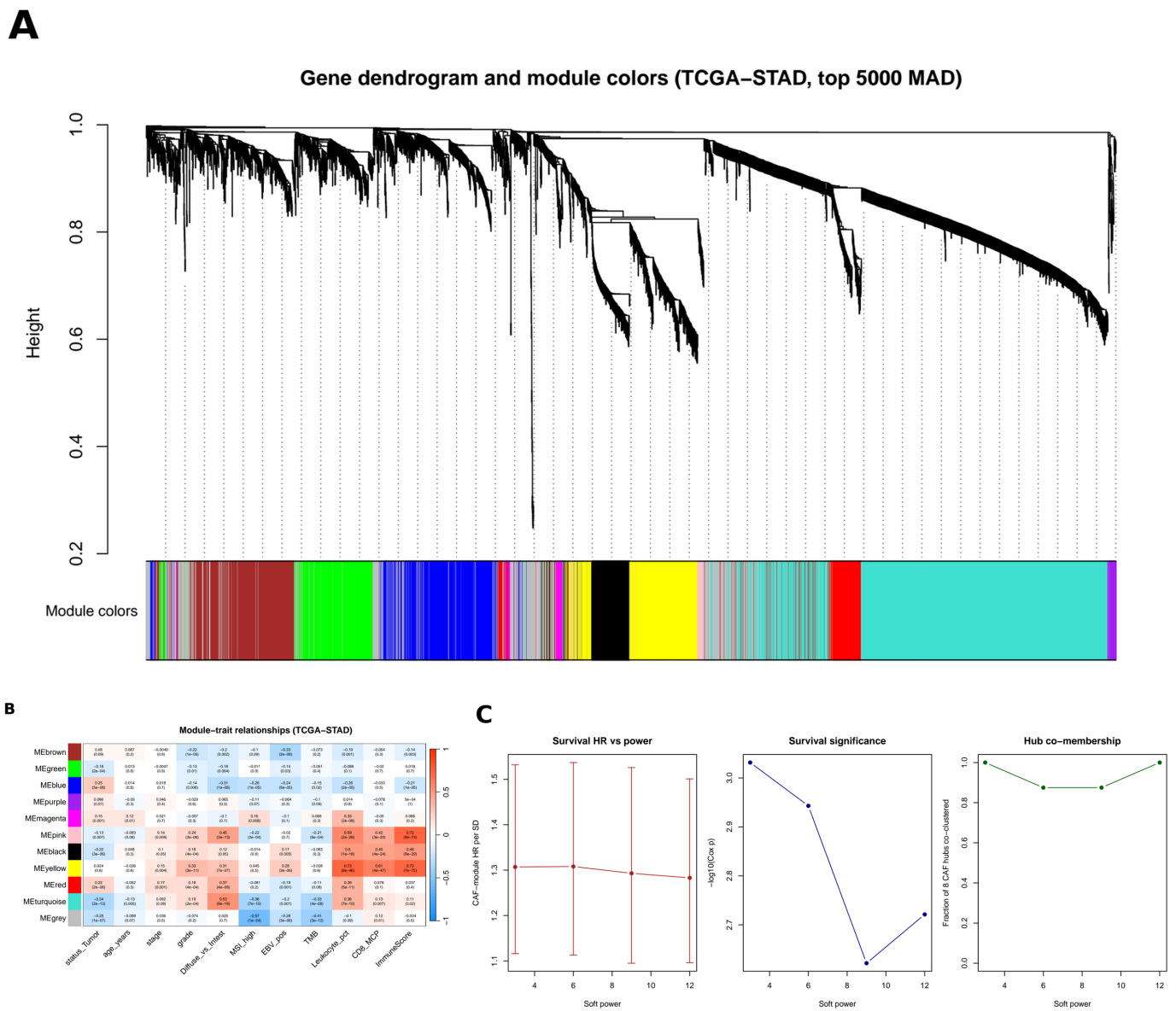
results/figures/Fig3.png

Figure 4



Immune microenvironment and the multi-cohort microbiome.
results/figures/Fig4.png

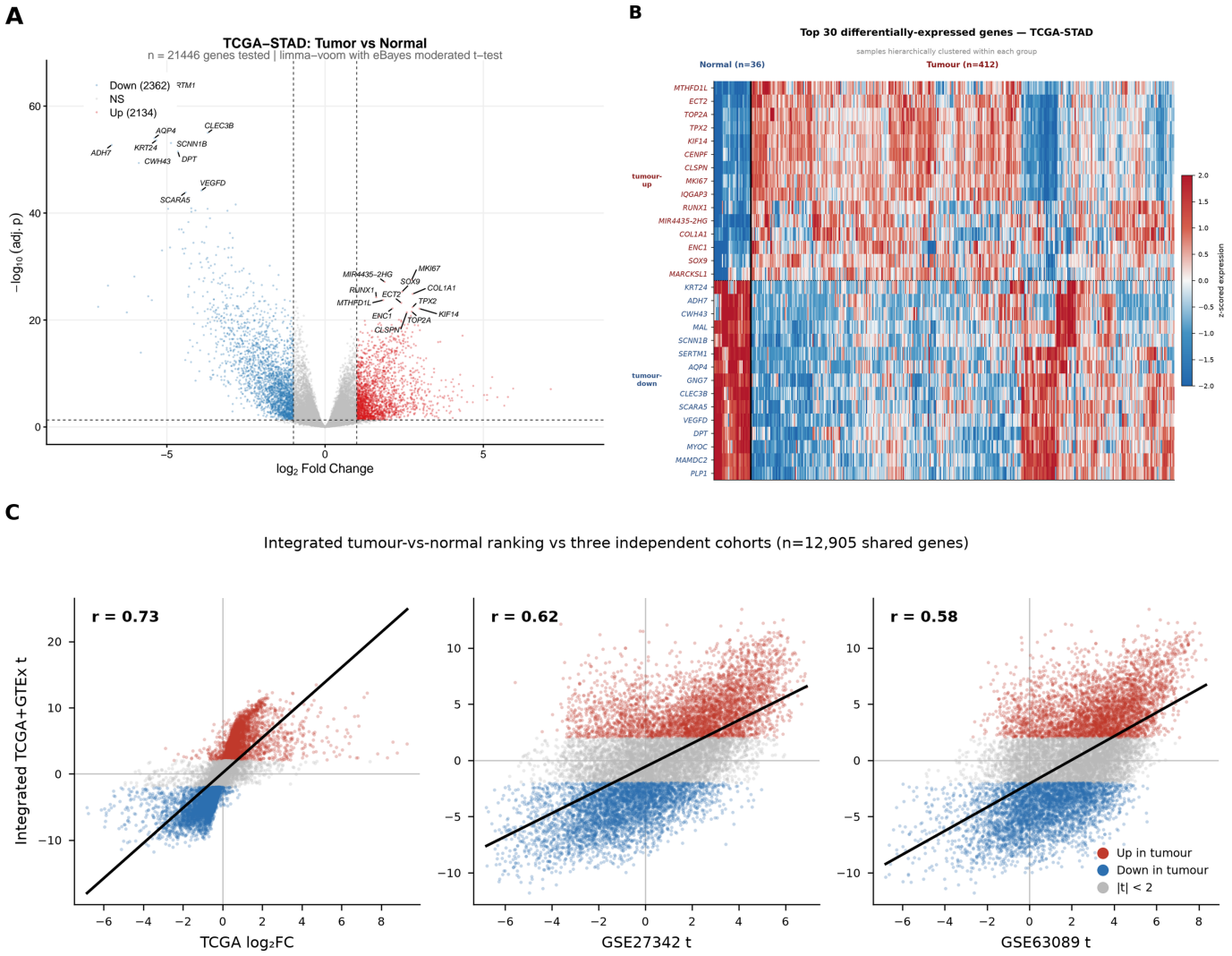
Figure 5



Weighted gene co-expression network analysis (WGCNA) of TCGA-STAD.

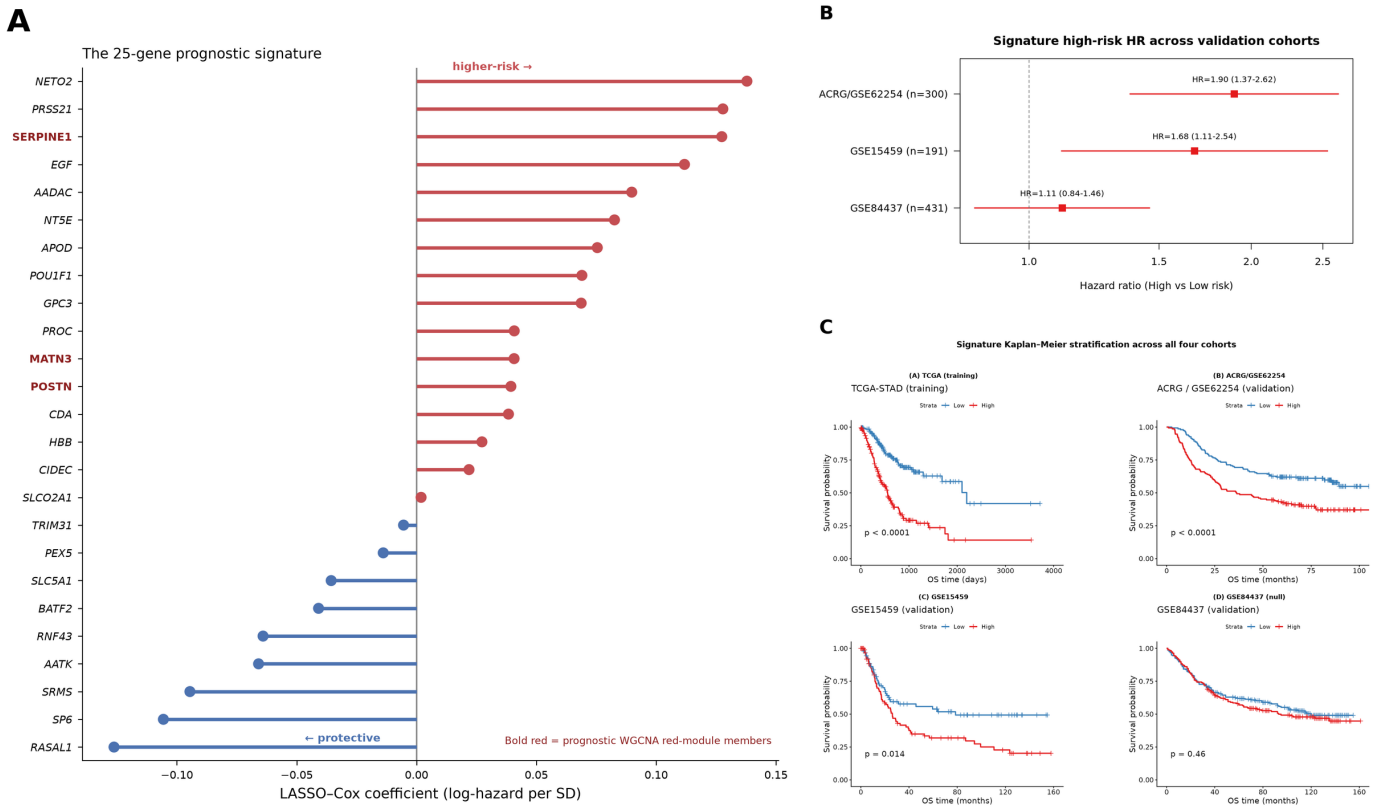
results/figures/fig5.png

Figure 6



TCGA-only differential expression.
results/figures/Fig6.png

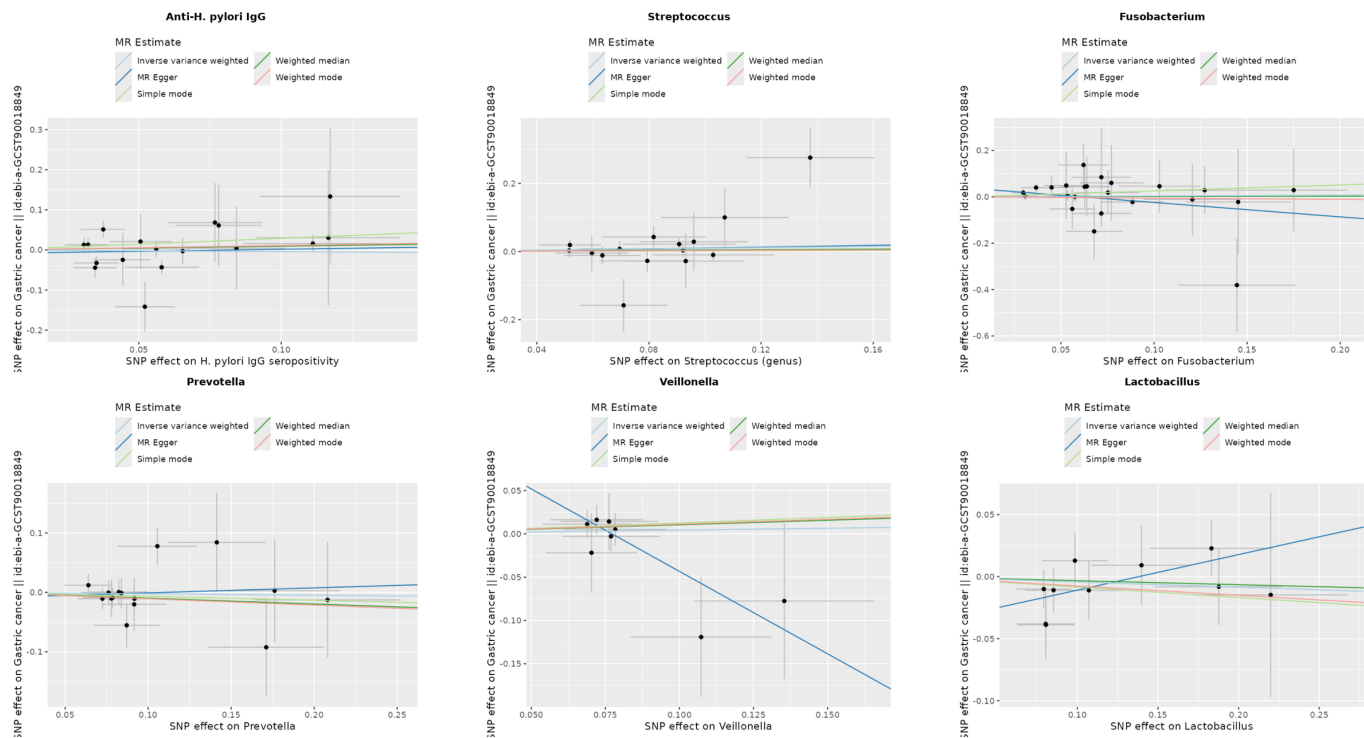
Figure 7



The 25-gene prognostic signature in detail.
results/figures/Fig7.png

Figure 8

Mendelian randomisation — SNP-effect scatter, all six exposures (European outcome)

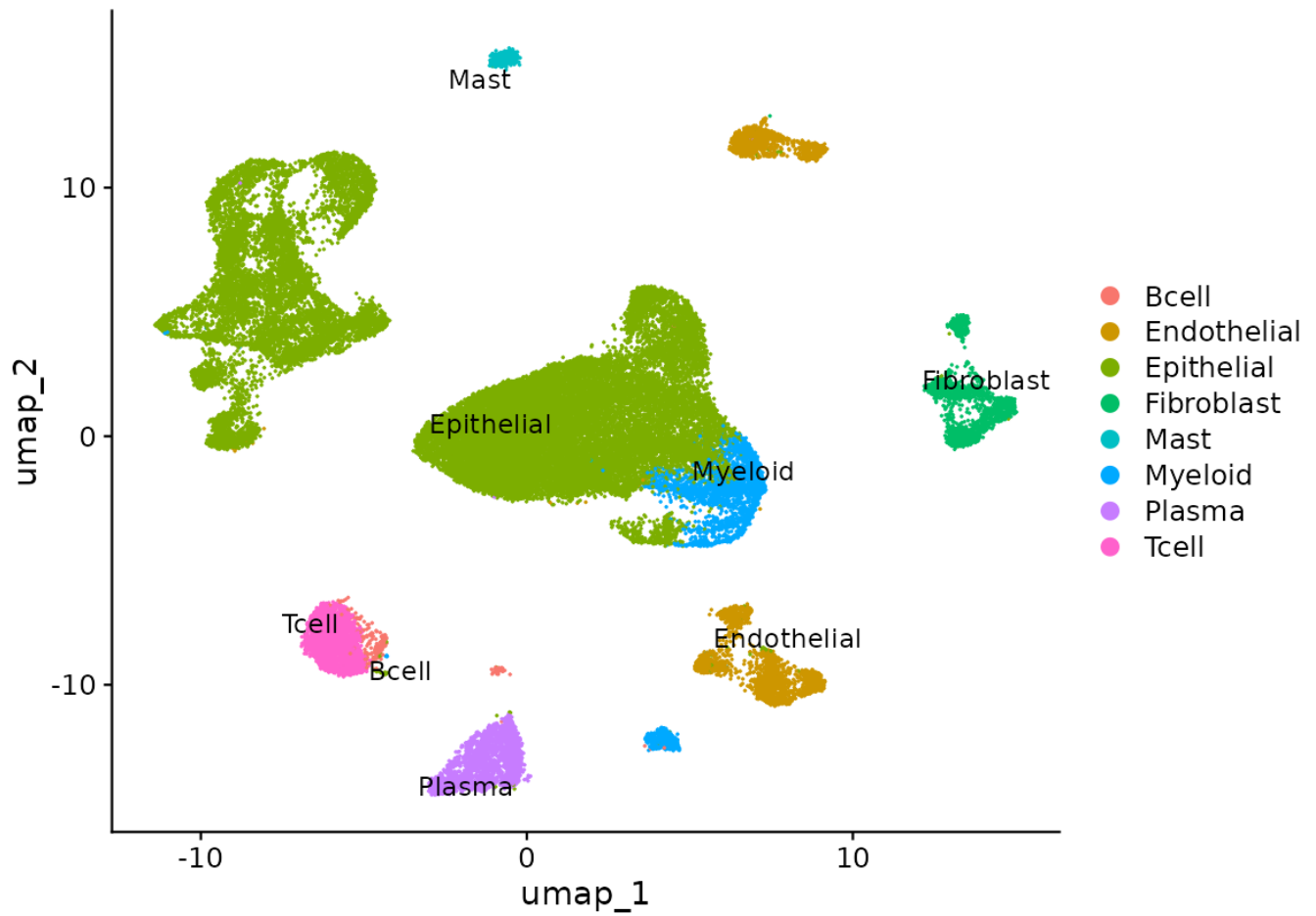


Mendelian-randomisation SNP-effect scatter, all six exposures.

results/figures/Fig8.png

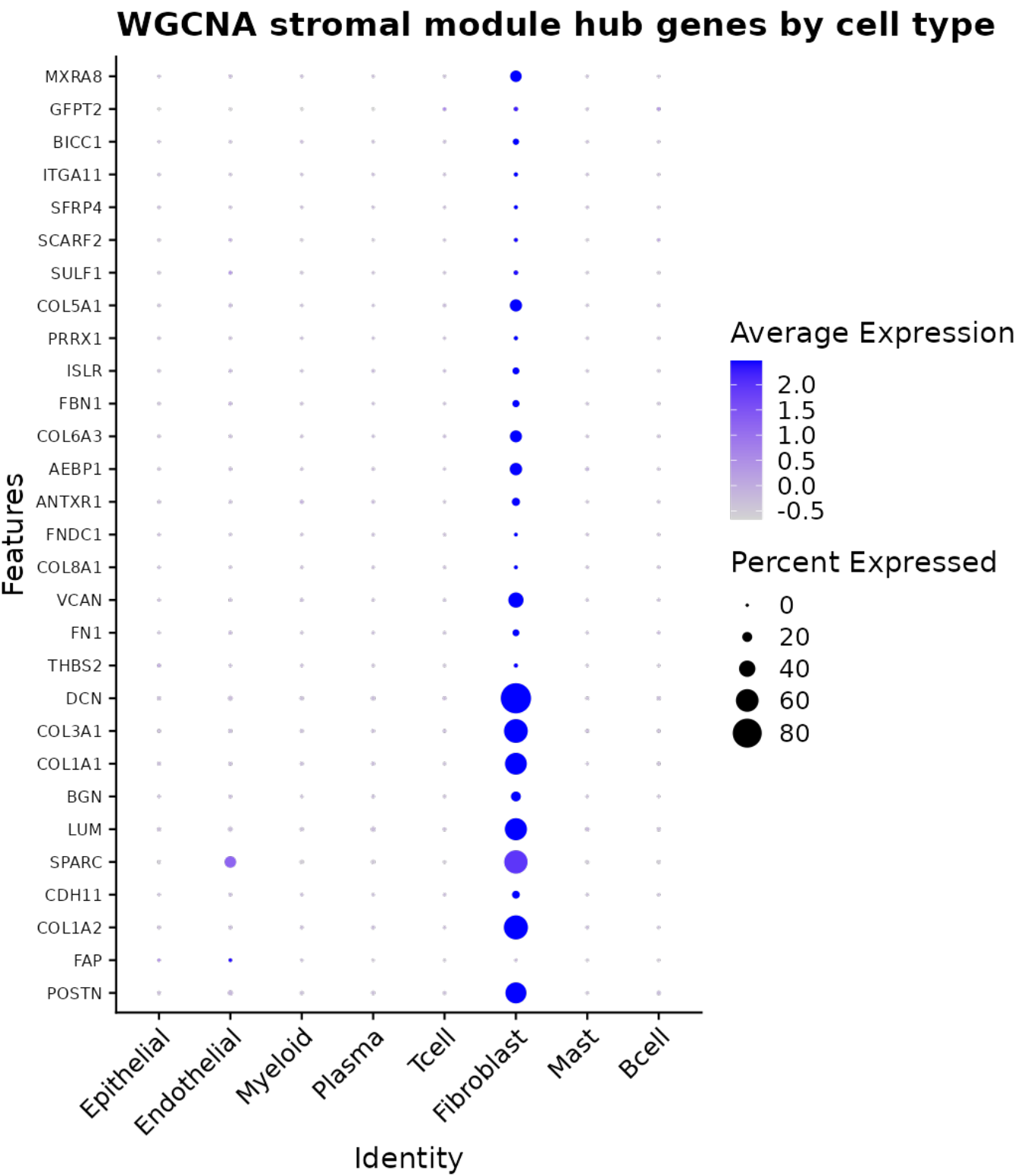
Supplementary Figure S1

GSE134520 gastric mucosa scRNA-seq — cell types



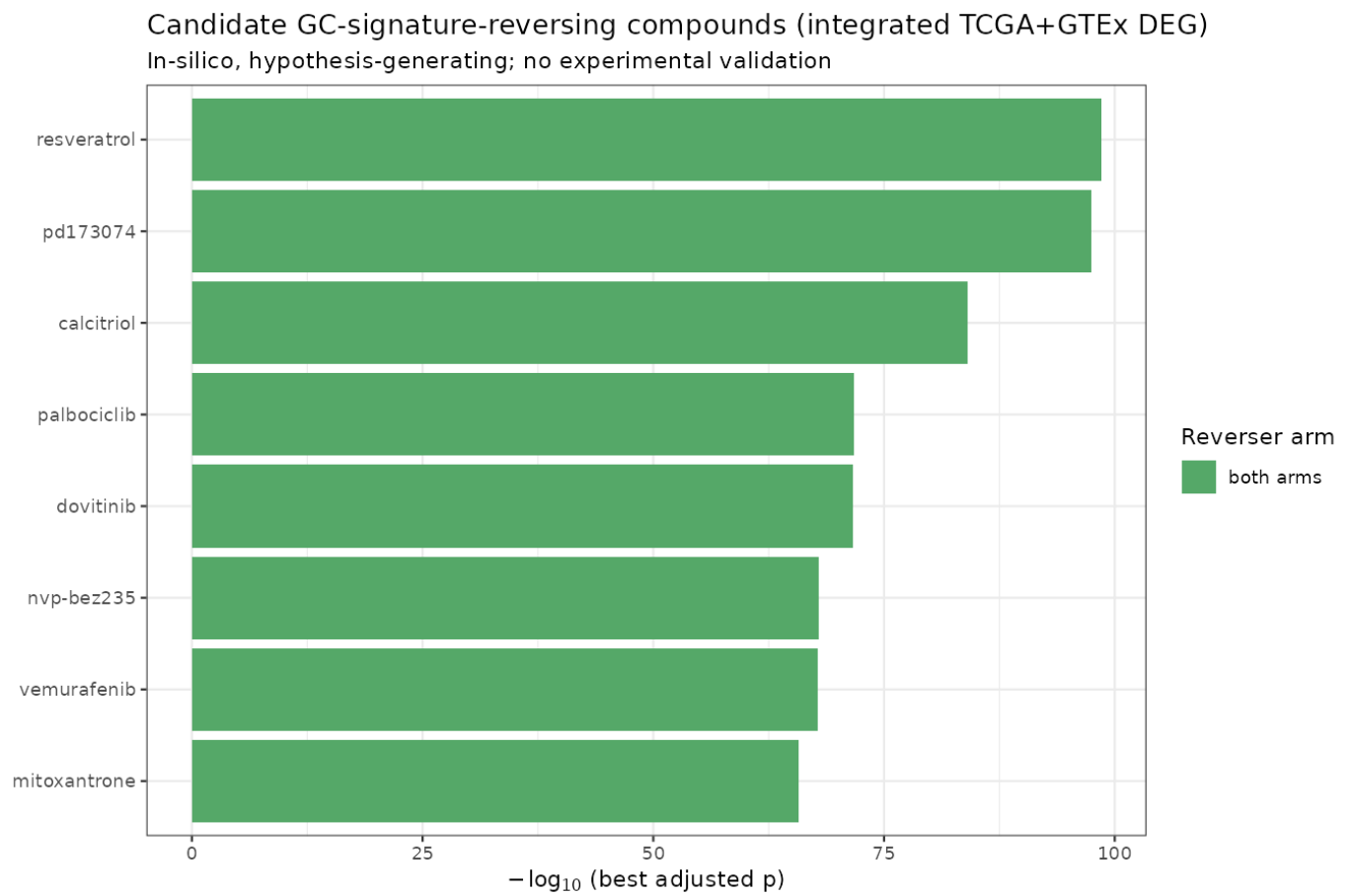
Single-cell atlas (GSE134520).
results/scrna/UMAP_celltypes.png

Supplementary Figure S2



Stromal hub-gene expression by cell type.
results/scrna/DotPlot_stromal_module_hub.png

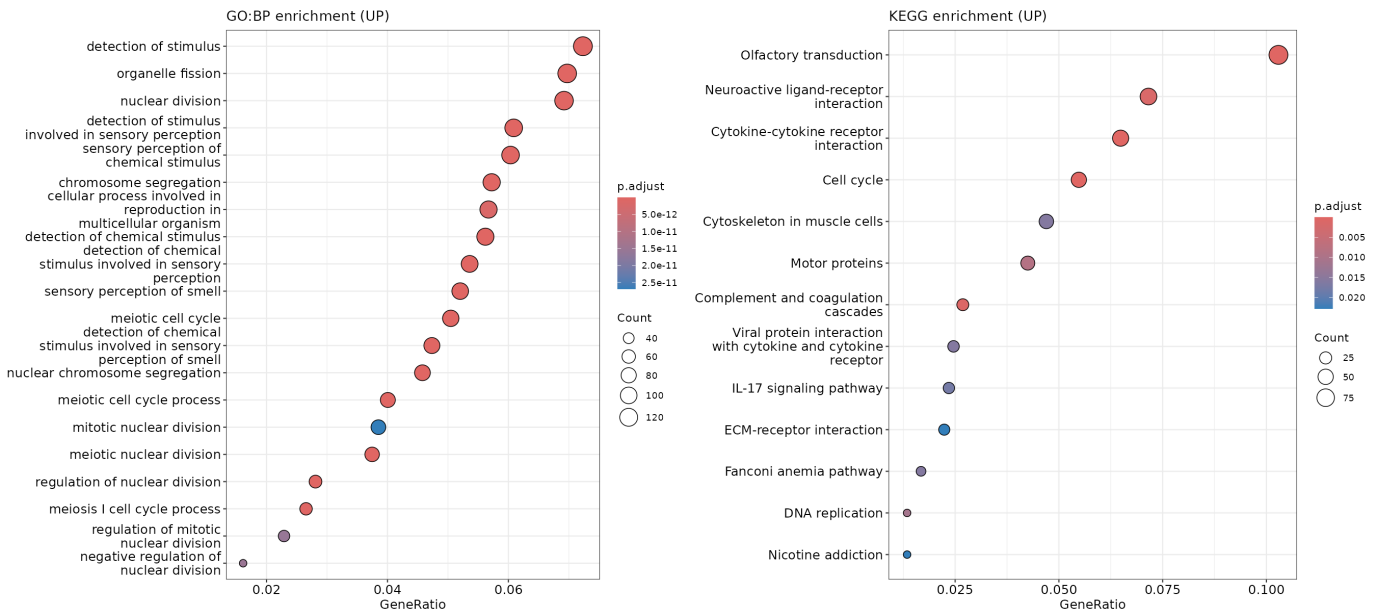
Supplementary Figure S3



In-silico drug-repurposing candidates.

results/drug_repurposing_integrated/top_candidate_drugs.png

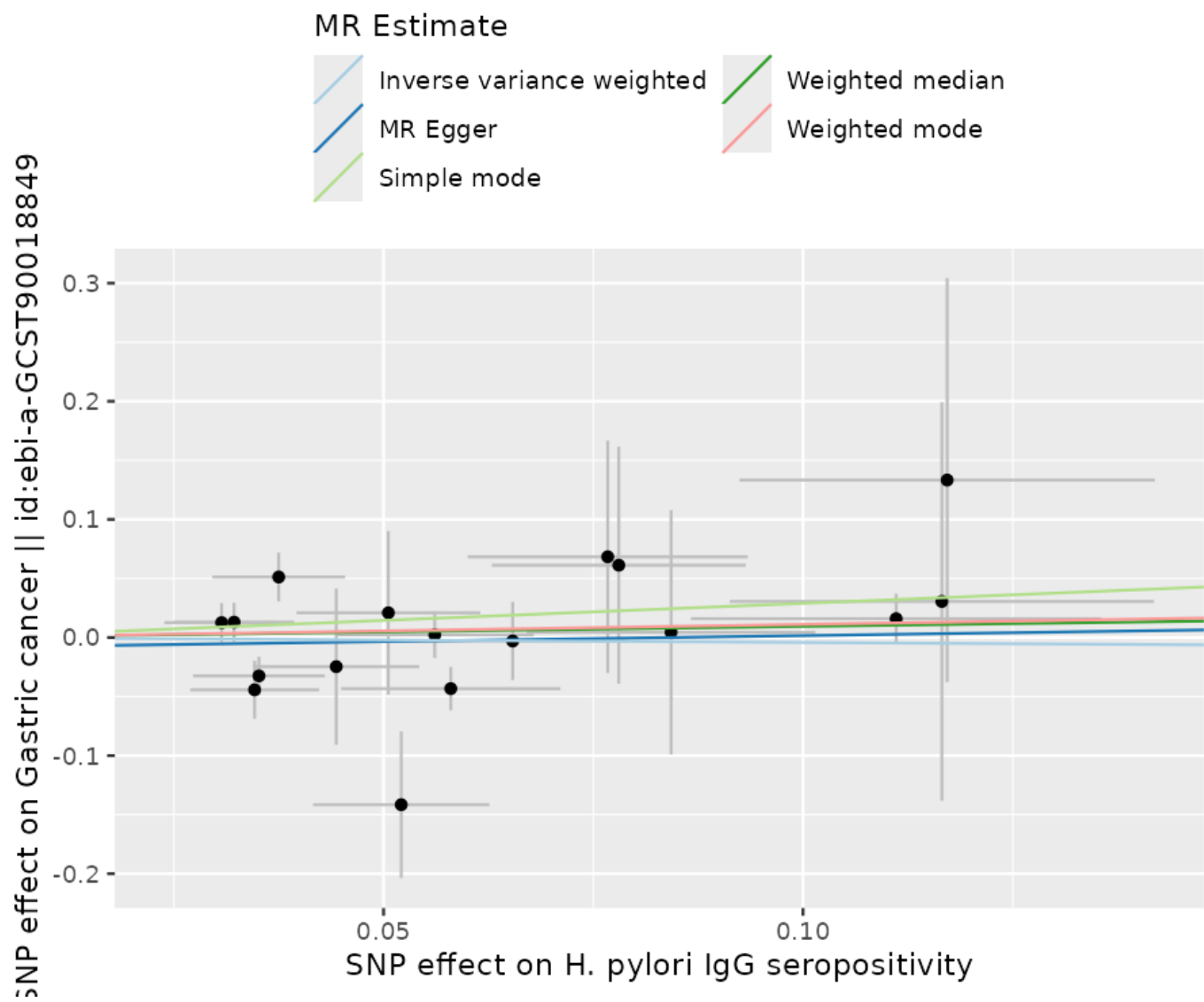
Supplementary Figure S4



Over-representation analysis (GO:BP and KEGG, tumour-up genes).

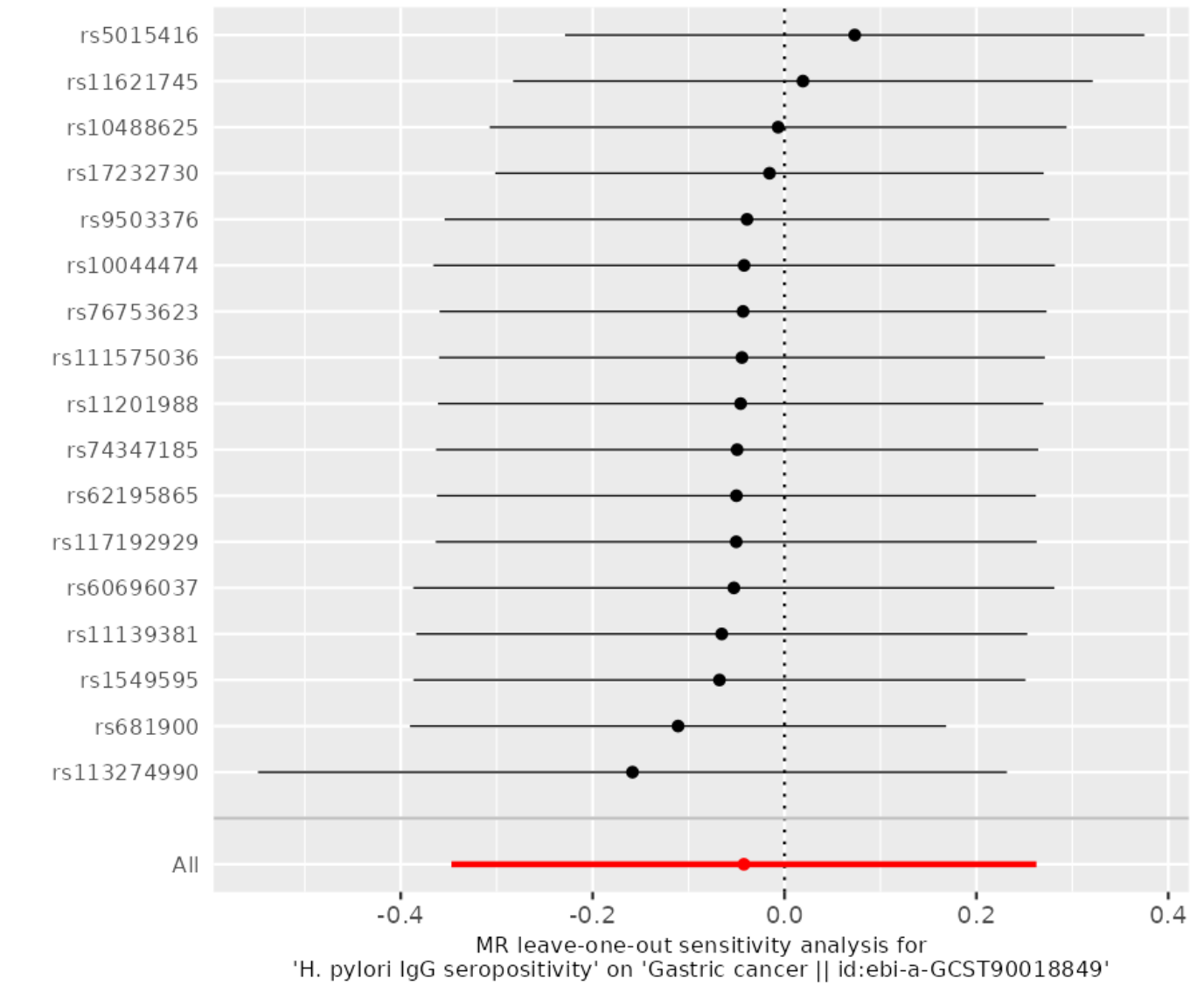
results/enrichment/path_ORA_GO_KEGG.png

Supplementary Figure S5



Mendelian-randomisation scatter (representative exposure).
results/mr_real/scatter_H_pylori_IgG_seropositivity.png

Supplementary Figure S6

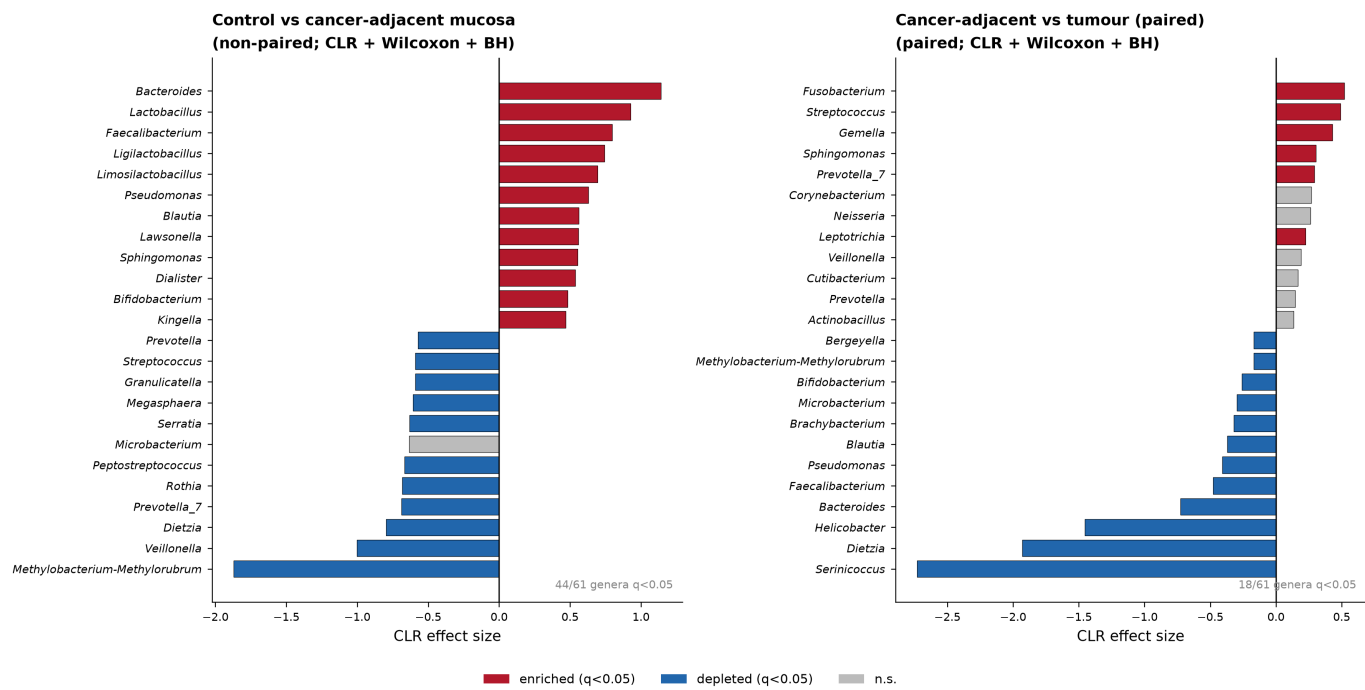


Mendelian-randomisation leave-one-out (representative exposure).

results/mr_real/loo_H_pylori_IgG_seropositivity.png

Supplementary Figure S7

Tissue-microbiome differential abundance (compositional; batch-cautioned)

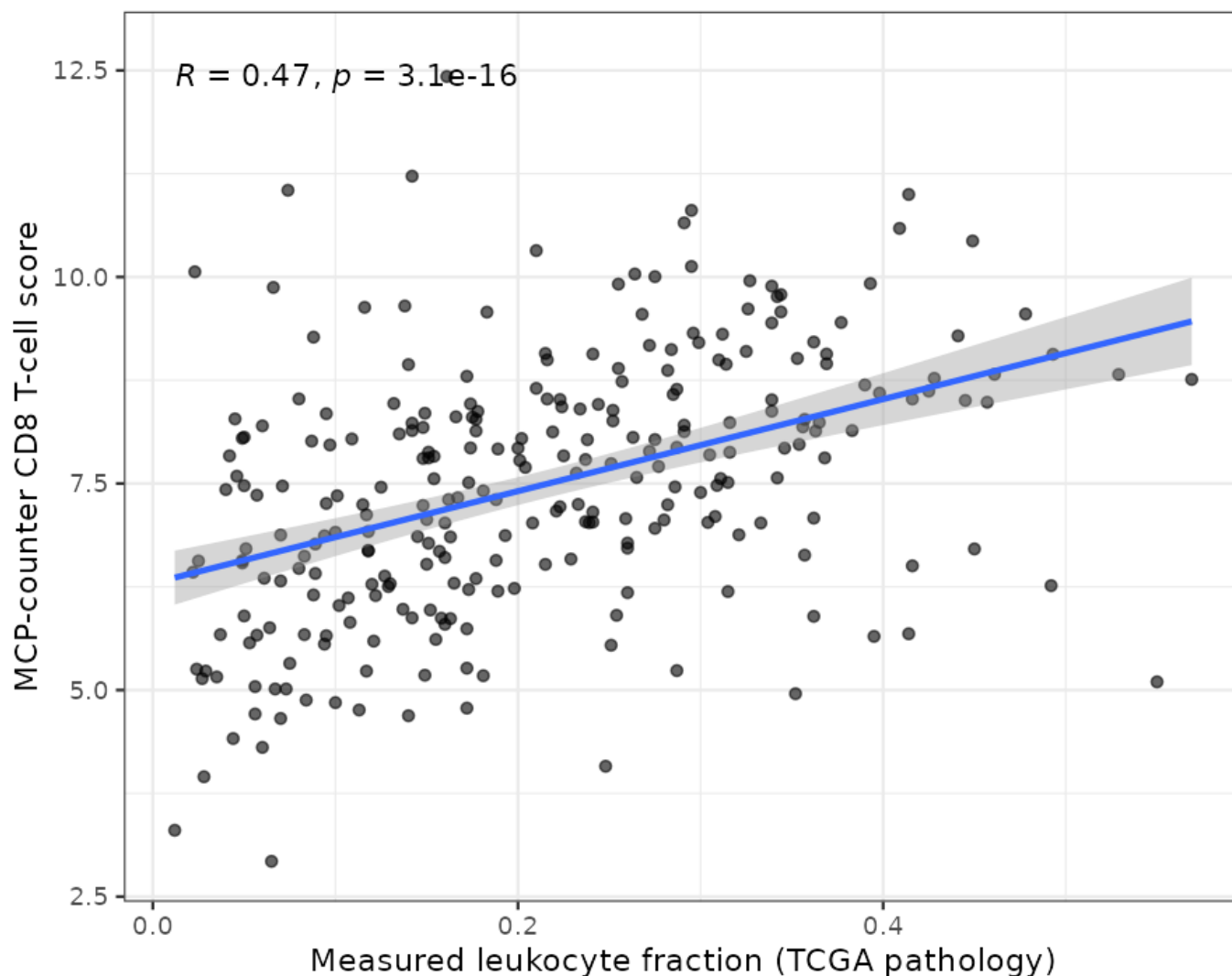


Tissue-microbiome compositional differential abundance.

results/microbiome_biomarker/da_clr_barplot.png

Supplementary Figure S8

MCP-counter CD8 T cells vs measured leukocyte % (tumors)

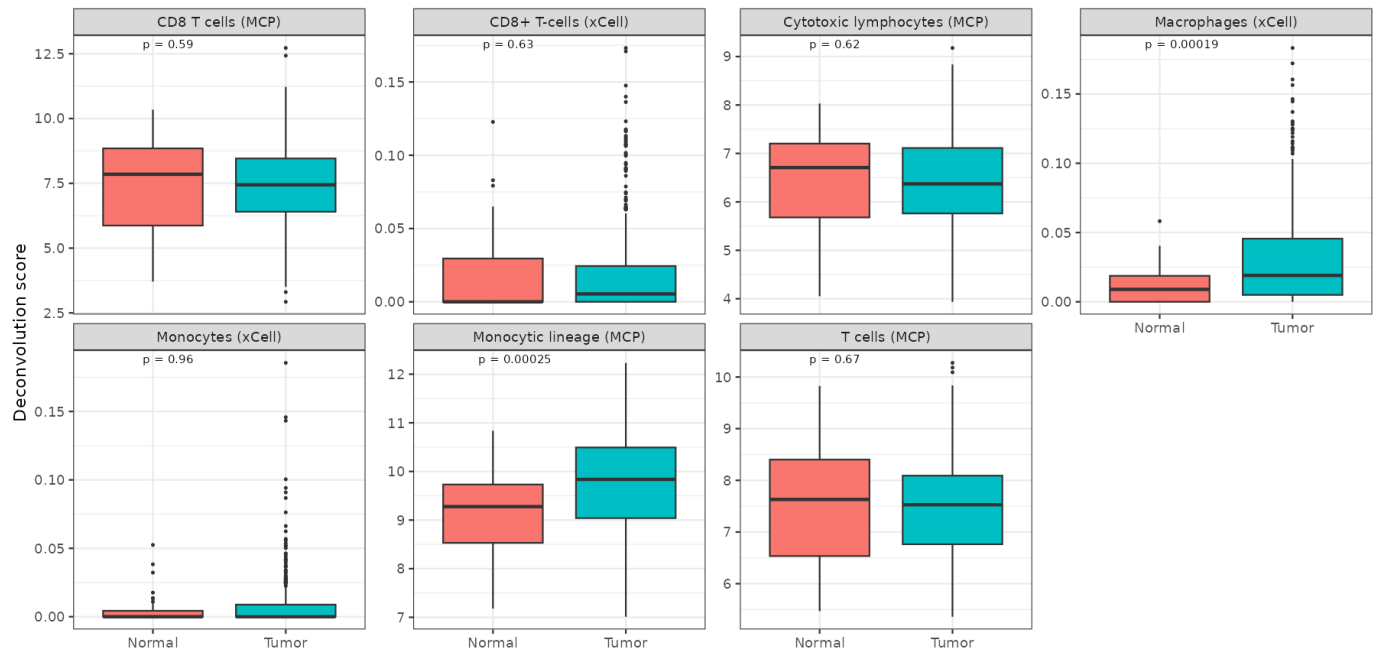


Deconvolution validated against pathology: MCP-counter CD8 T-cell score versus measured leukocyte percentage (Spearman $\rho = 0.468$, $p = 3.1e-16$, $n = 272$). The total T-cell score correlates more strongly still ($\rho = 0.666$, $p = 3.6e-36$). Neither estimate correlates with measured lymphocyte-infiltration percentage (CD8 $\rho = 0.020$, $p = 0.74$), so deconvolution captures overall immune burden rather than lymphocyte subset composition.

[results/plots/Immune_validation_scatter.png](#)

Supplementary Figure S9

Immune populations: Tumor vs Normal (TCGA-STAD)

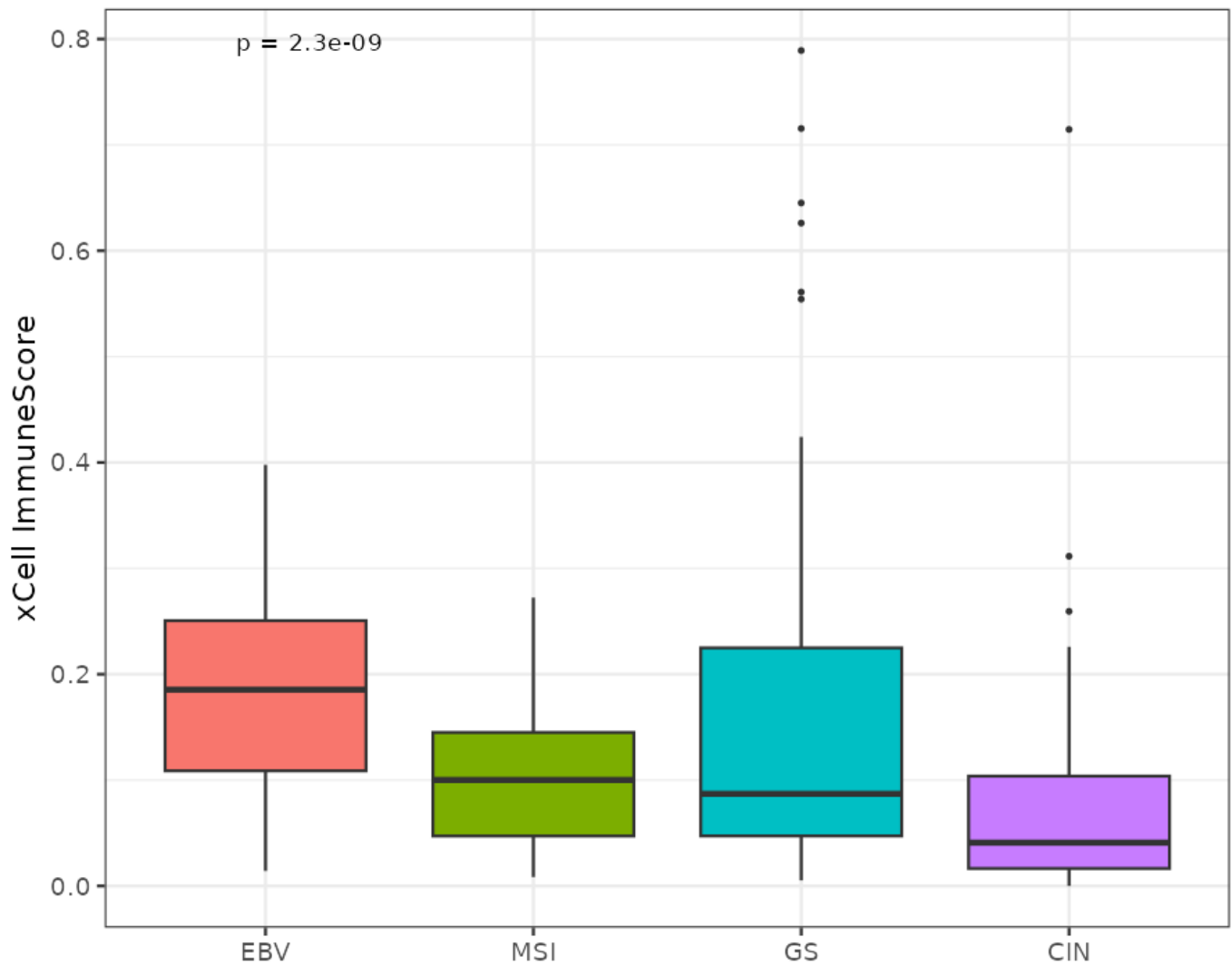


Immune cell-type estimates, tumour versus normal tissue.

results/plots/Immune_tumor_vs_normal.png

Supplementary Figure S10

xCell ImmuneScore by molecular subtype (TCGA-STAD)

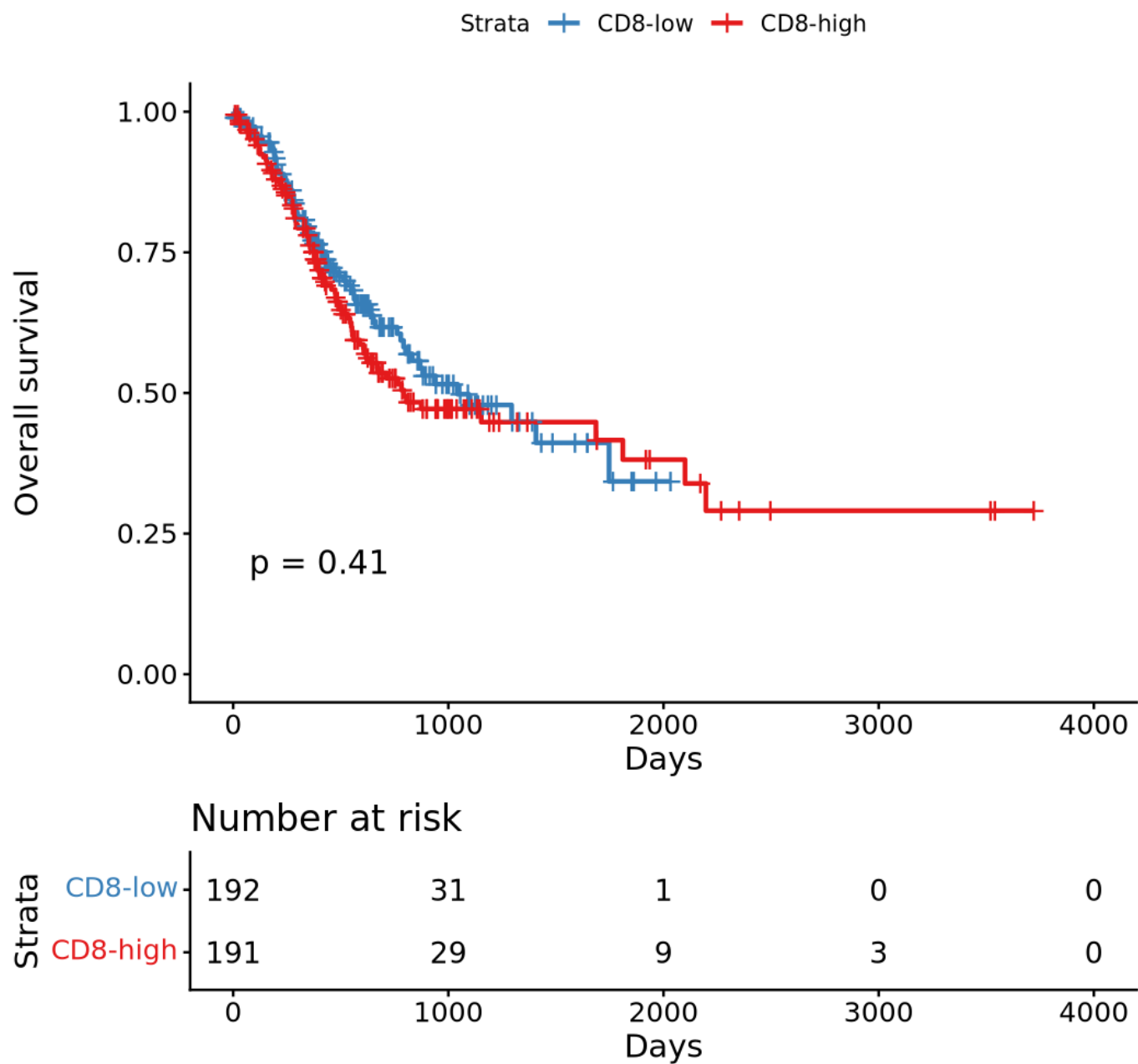


Immune infiltration by Lauren subtype.

results/plots/Immune_by_subtype.png

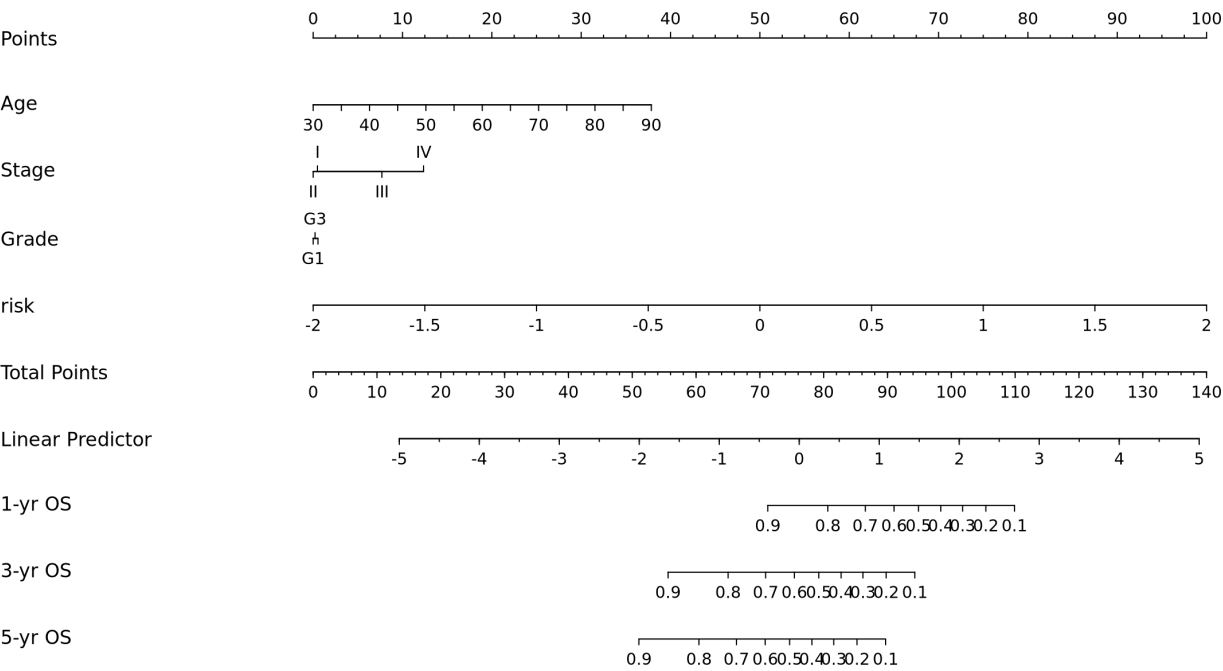
Supplementary Figure S11

OS by MCP-counter CD8 T-cell score (median split)



CD8 T-cell infiltration and overall survival (HR 1.04, $p = 0.41$; not prognostic).
results/plots/Immune_CD8_survival_KM.png

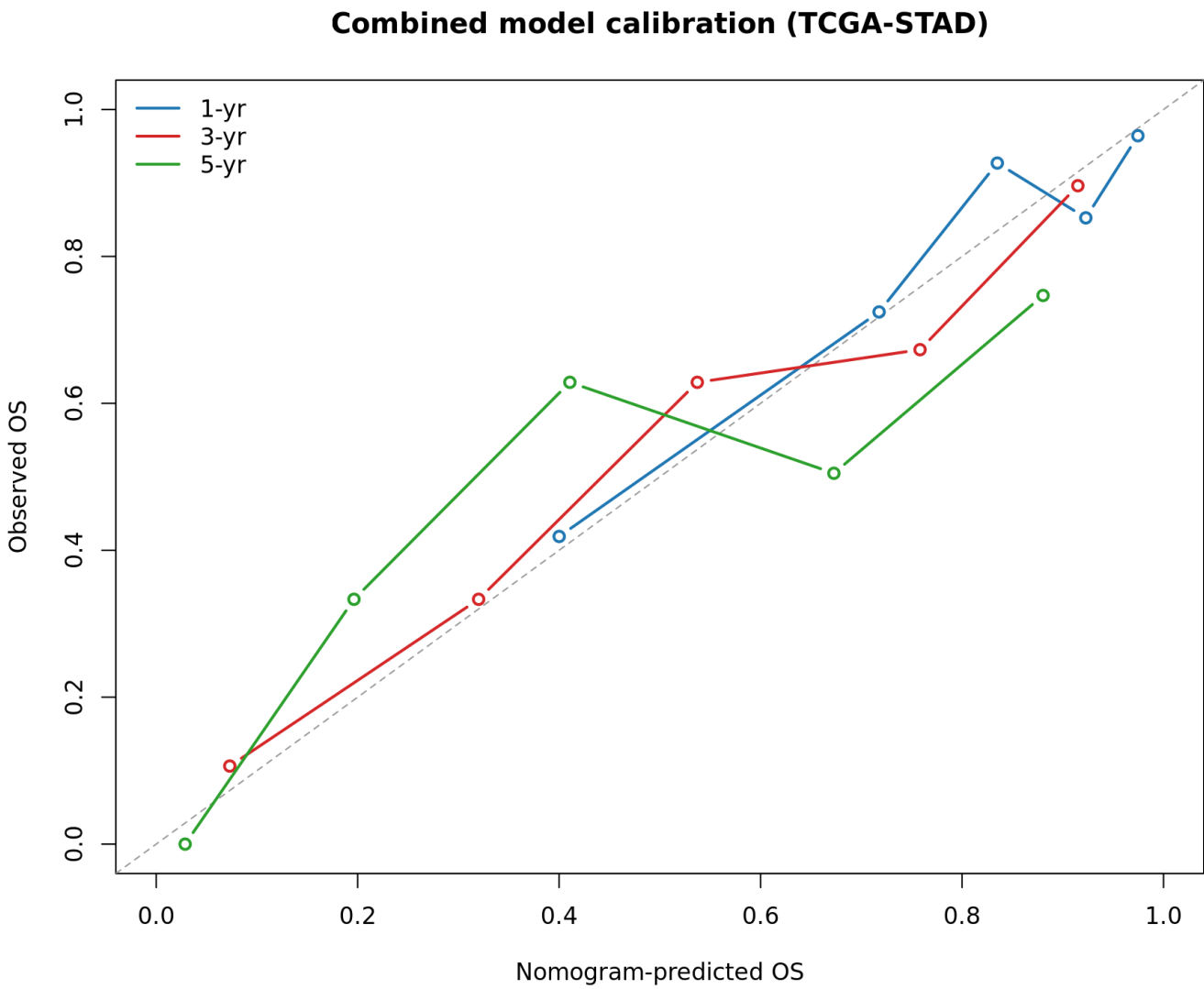
Supplementary Figure S12



Clinical nomogram combining stage, age and the 25-gene signature.

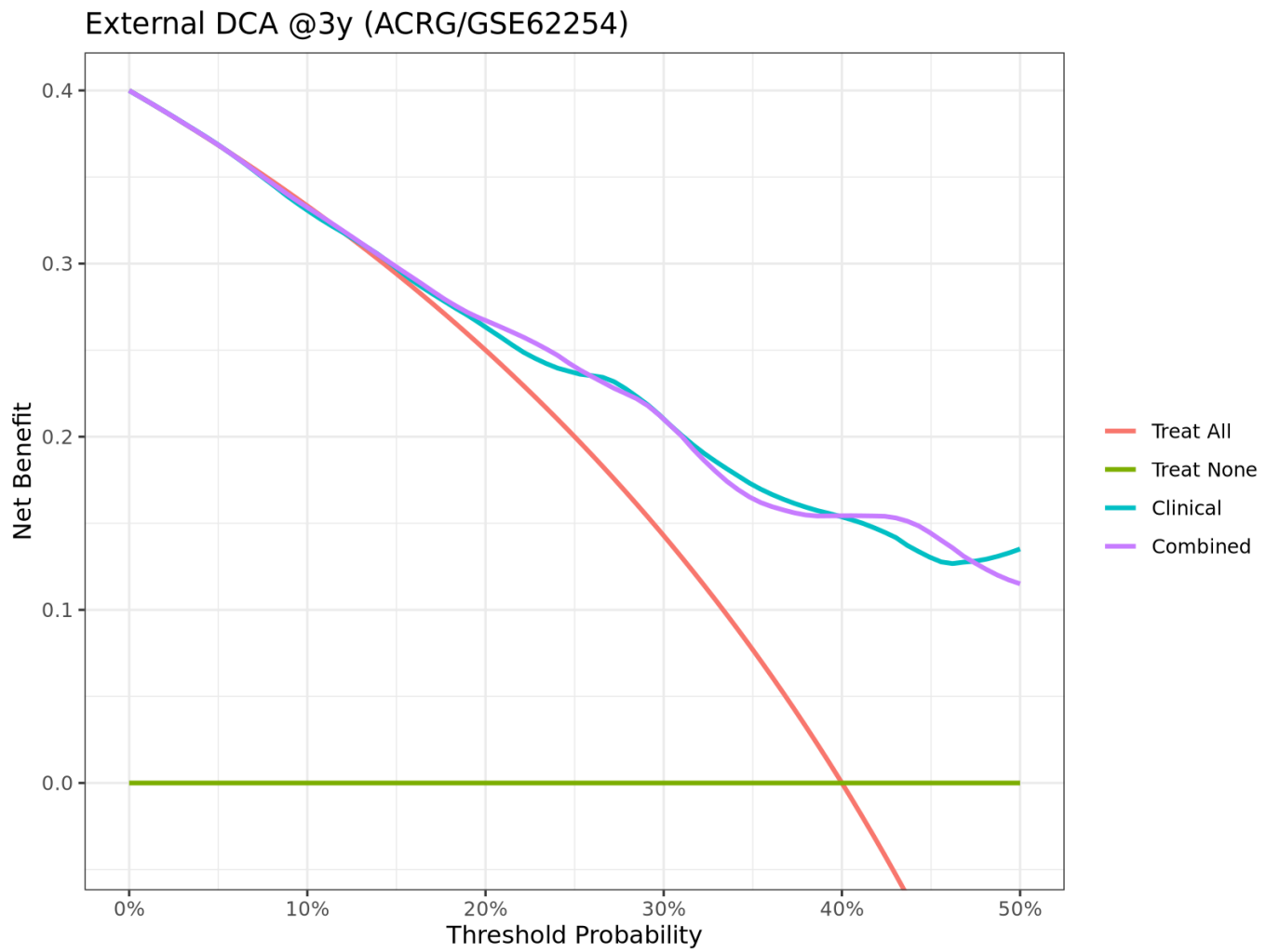
[results/nomogram_combined/combined_nomogram.png](#)

Supplementary Figure S13



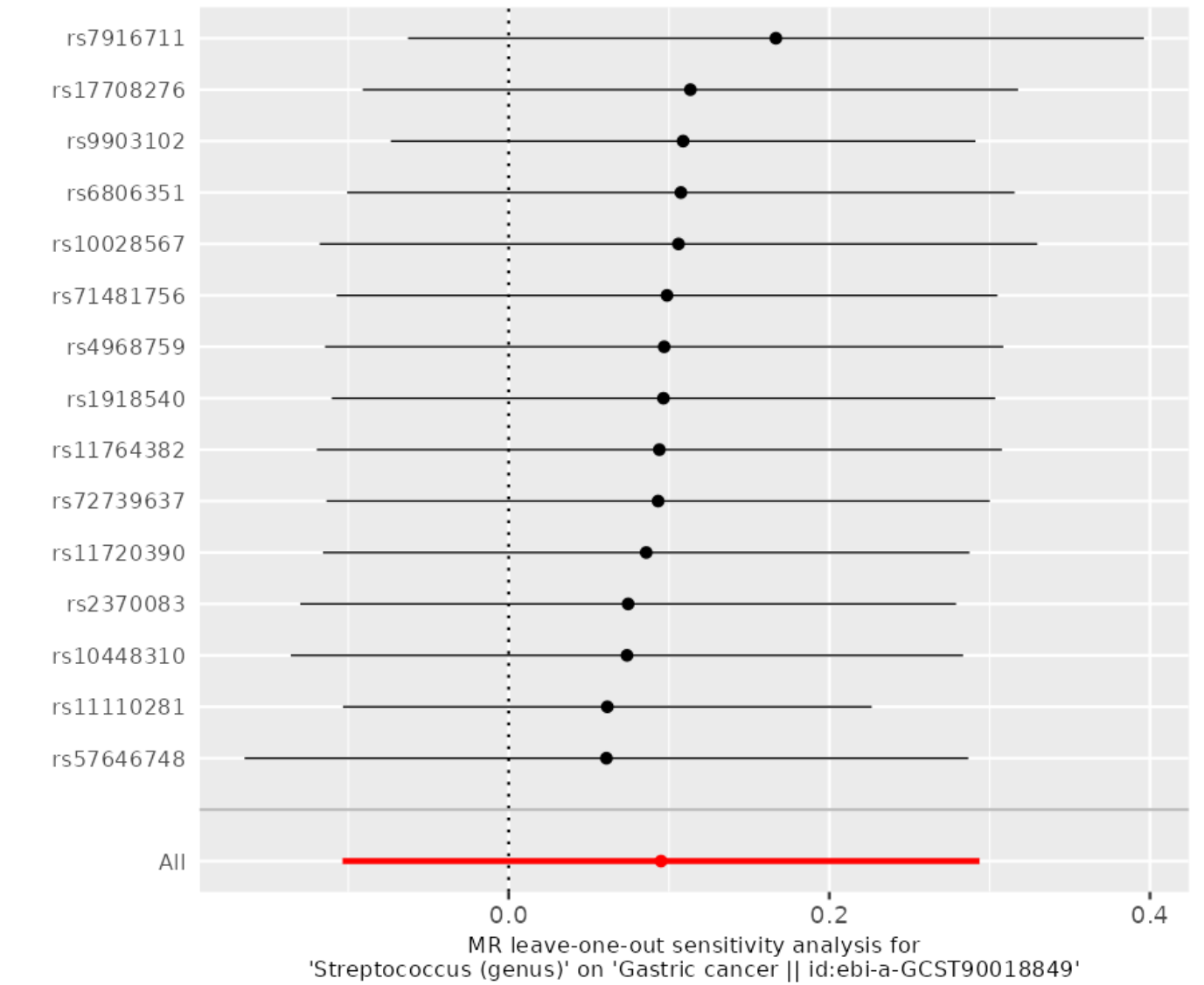
Calibration of the combined model against observed survival.
results/nomogram_combined/calibration_combined.png

Supplementary Figure S14



External decision-curve analysis in ACRG: no incremental net benefit over staging (delta-C +0.005).
results/external_utility_ACRG/DCA_external.png

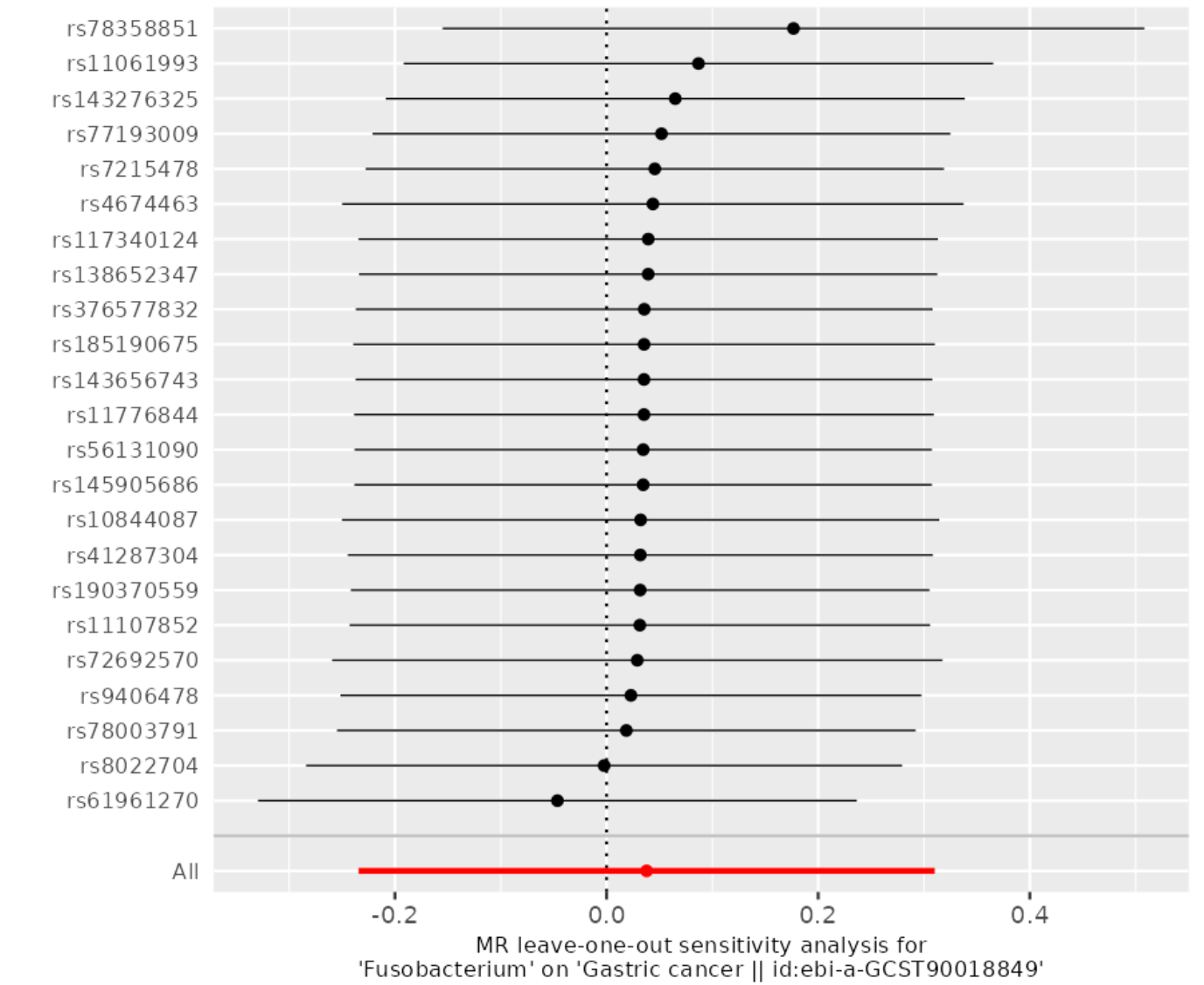
Supplementary Figure S15



Mendelian-randomisation leave-one-out: Streptococcus (genus).

results/mr_real/loo_Streptococcus__genus_.png

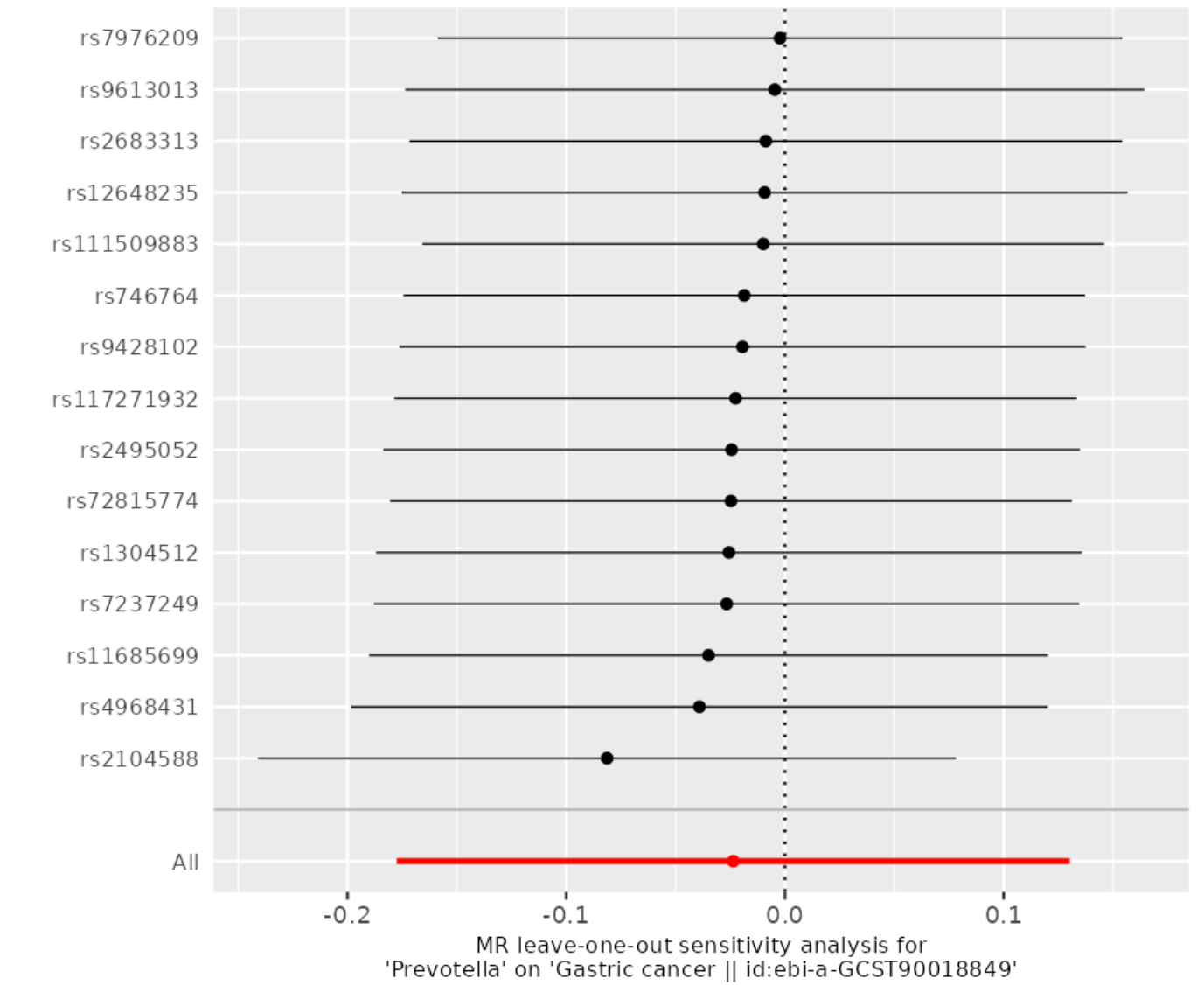
Supplementary Figure S16



Mendelian-randomisation leave-one-out: Fusobacterium.

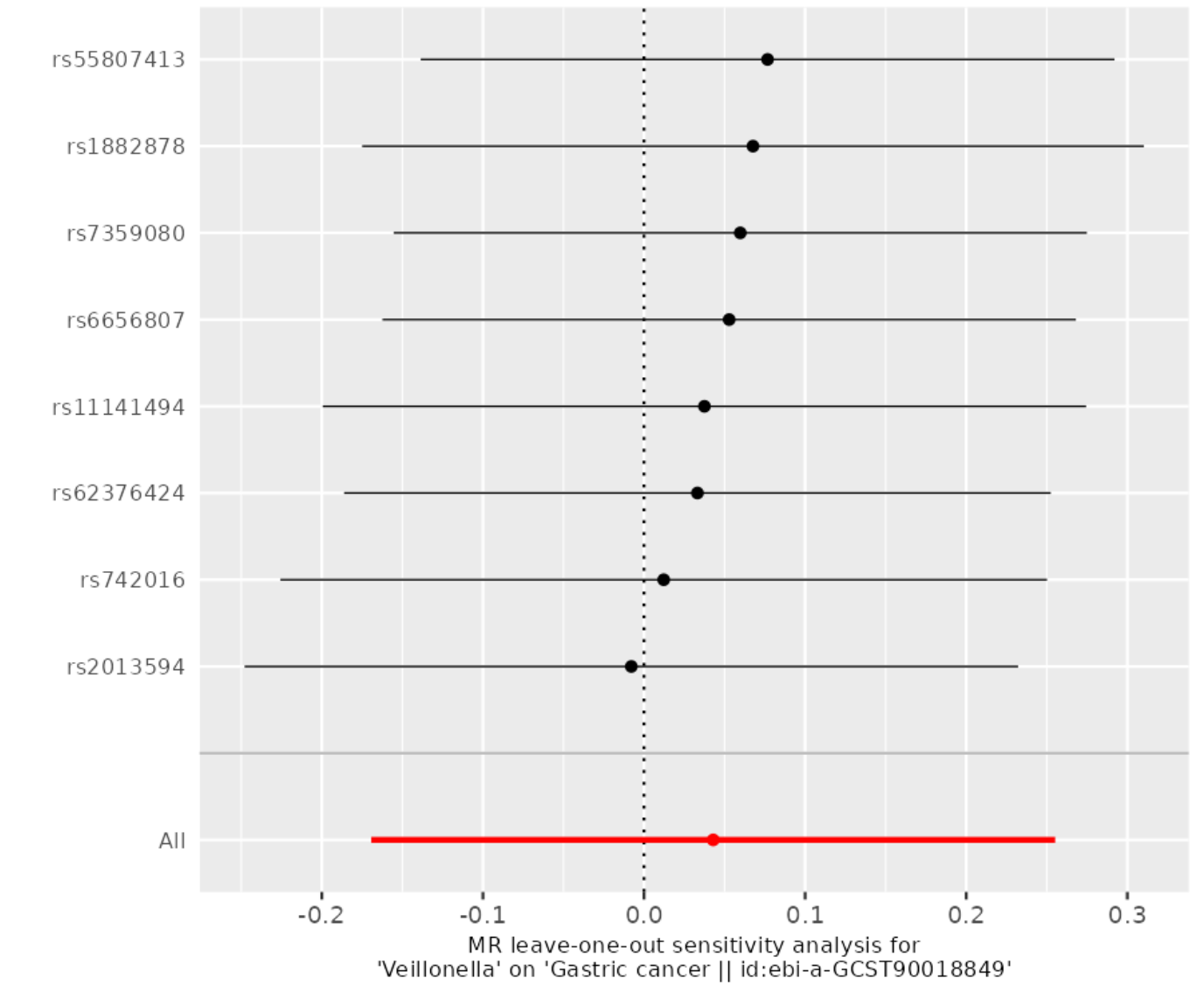
results/mr_real/loo_Fusobacterium.png

Supplementary Figure S17



Mendelian-randomisation leave-one-out: Prevotella.
results/mr_real/loo_Prevotella.png

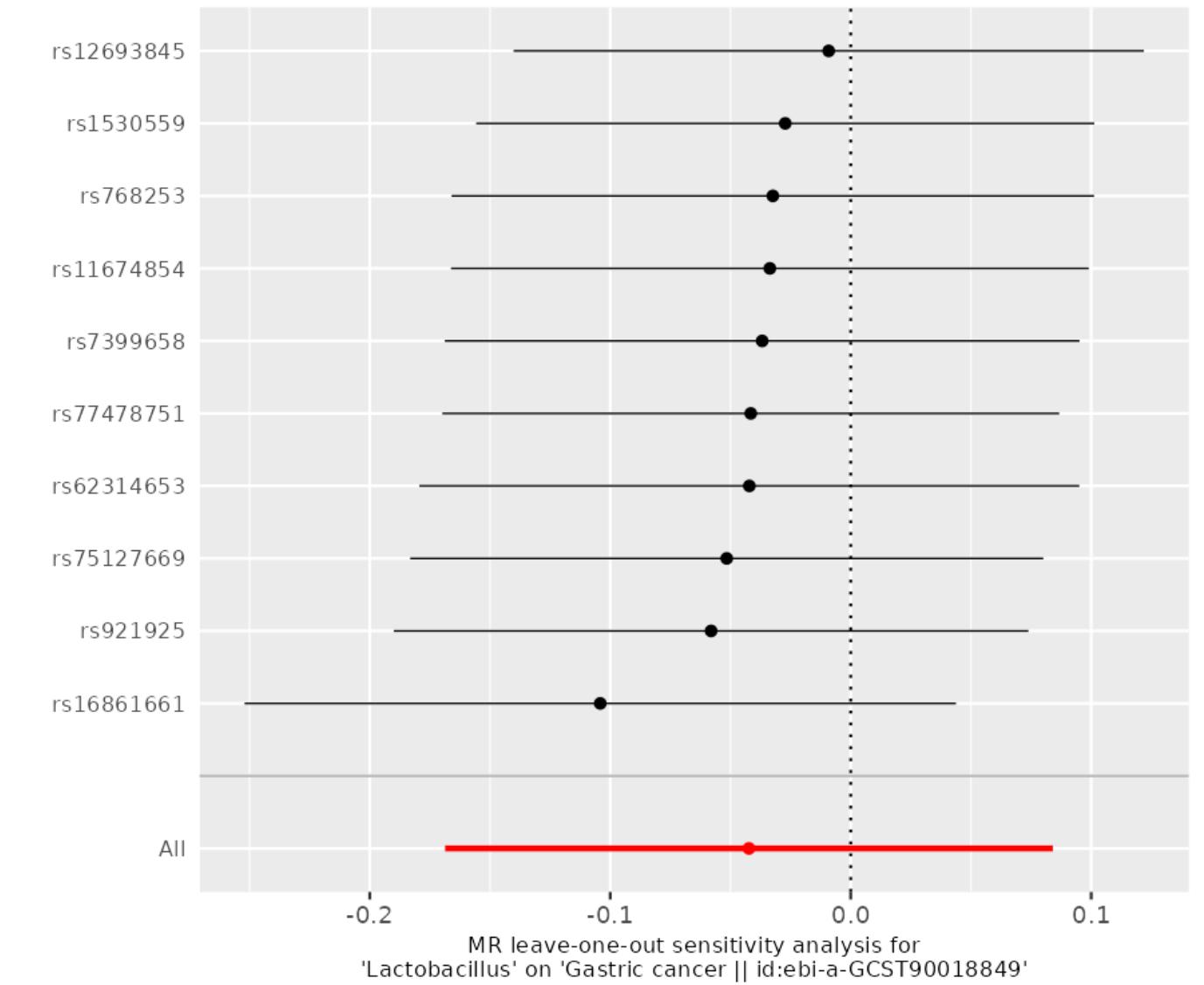
Supplementary Figure S18



Mendelian-randomisation leave-one-out: Veillonella.

results/mr_real/loo_Veillonella.png

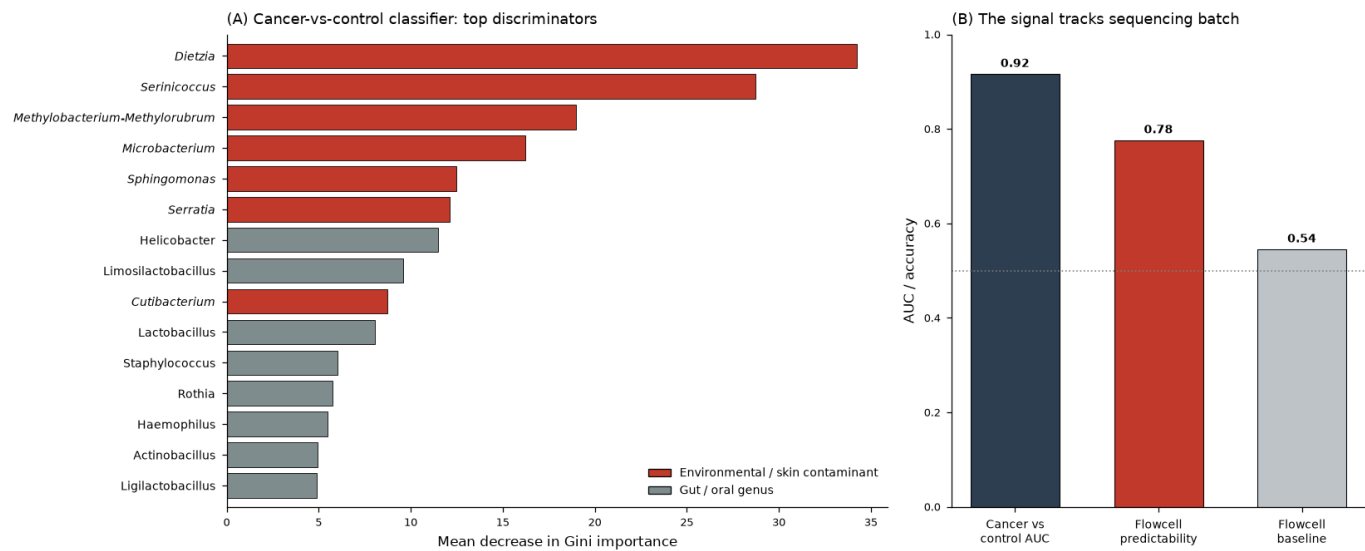
Supplementary Figure S19



Mendelian-randomisation leave-one-out: Lactobacillus.
results/mr_real/loo_Lactobacillus.png

Supplementary Figure S20

Tumour-microbiome random forest: apparent accuracy is a batch artefact



The tumour-microbiome classifier is a batch artefact.

results/microbiome_biomarker/rf_batch_classifier.png